



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Metody uczenia reprezentacji dla generalizacji modeli świata i polityk z symulacji do rzeczywistości w fizycznej sztucznej inteligencji / Representation learning methods for the simulation-to-reality generalization of world models and policies in physical AI

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	Doctoral School courses. Reading the literature and considering possible research directions
Semestr 2	Doctoral School courses and literature review. Choosing the dissertation topic with the supervisor and preparing the Individual Research Plan
Semestr 3	Setting up the research environment with world models, public real robot datasets and the SIMPLER simulator. Initial experiments on how the prediction target of a world model and the encoder of the policy affect the performance on real data of policies trained in it (RQ1). Starting to build the pipeline for simulation data and its photorealistic transfer (RQ2)
Semestr 4	Further experiments on RQ1 in manipulation. Preparing and submitting the first publication to a 200-point venue in machine learning or robot learning, such as ICLR or IEEE RA-L. First experiments on world models trained on simulation data (RQ2)
Semestr 5	Experiments on world models trained on simulation data that transfer to real data (RQ2) and first experiments on pretraining that reduces the real data needed for adaptation (RQ3). Preparing a scientific publication
Semestr 6	Experiments on reducing the real data needed for adaptation (RQ3) and on transfer to unseen scenes and, for RQ1, to navigation (RQ4). Preparing a publication and, if access allows, tests on a real robot
Semestr 7	Final experiments and submitting a paper to a conference or journal. First draft of the dissertation
Semestr 8	Completing and revising the doctoral dissertation, then submitting it



4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

September 2029

(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Robots that learn by trial and error need far more attempts than a real robot can make safely or cheaply, so they are usually trained in simulation. Hand-built physics simulators are fast, but building a realistic copy of each new place takes a lot of work, and their images still look different from real camera images. A newer option is a learned world model [1]. It is a neural network that predicts what the robot will see after it takes an action. Recent world models are trained on large amounts of real video and can serve as simulators. UniSim [2] trained robot policies only inside such a model and then used them on real robots. Navigation World Models [3] imagine how a walk through an unfamiliar place would look from a single image of it, and NVIDIA's Cosmos models [4] are released as a base for building such simulators.

This gives a loop from reality to simulation and back. Real recordings train a world model, a policy is trained inside the world model, and the policy is then used in the real world. The weak point is that an imagined rollout is never exact. The model may ignore an action or predict motion that breaks physics [5],[6], and small errors add up over a long rollout. A policy that succeeds in the world model can therefore fail on a real robot. Better-looking video does not solve this, because the visual quality of a world model does not reliably predict whether policies succeed in it [6]. Two further things are open. Physics simulation data are cheap and unlimited, but a world action model built on a video model and trained only on demonstrations from a physics simulator succeeded in about a third of the trials on a real robot arm in the first such study I have found [7]. I have not found in my search a controlled comparison of ways to build a world model from such data so that it predicts the real world well. And when the world model or the policy meets a new real setting, I have not found a settled answer to how to adapt it with as few real samples as possible.

I want to approach these three problems through the representations the models learn. A neural network does not act on the image directly. It first turns it into internal features, called representations, and decides from them. How these features are learned is a choice: what the world model is trained to predict, which data and which encoder it starts from, and which objectives push the features to ignore what differs between a simulated and a real image. LeCun argued that a world model should predict in such a representation space rather than in pixels, so that it can ignore details that cannot be predicted and do not matter for control [8]. V-JEPA 2 [9] showed that a world model of this kind can plan the movements of a real robot arm, and SkyJEPA [10] trained one on simulated



states without real flight data and flew a real quadrotor with it. My working assumption follows the same idea. Features that leave out what differs between simulated and real images should carry a policy across the gap and should need less real data to adapt. I will judge this by performance on real data, measured on recorded real robot data and in SIMPLER, a simulator of scenes rebuilt from real recordings, and on a real robot only if access allows. I will study three connected questions: which representation learning choices affect how well a policy trained in a world model built from real data performs on real data, how a world model trained on physics simulation data should be built so that it predicts the real world, and how the representations of the world model and of the policy should be pretrained so that a new real setting needs few real samples. I will study robotic manipulation as the main task and check the main results in unseen scenes and in robot navigation.

The answers would show which properties of a learned simulator must be right for a given task and how to get a robot working in a new place with little real data. In practice, a robot could learn a new job in a warehouse, a hospital or a home mostly from simulated and predicted experience and then work there reliably after a short real-world adaptation. The same questions arise in autonomous driving, where generative world models already produce scenarios for training and testing, and in World Action Models, which predict the future and choose actions within one network.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The idea of learning a model of the environment and training a controller inside its predictions goes back at least to Ha and Schmidhuber [1], whose agent learned to play a game inside its own "dream" and then played the real game. DreamerV3 [11] turned training in imagination into a general method that works with one fixed configuration across more than 150 tasks. Such models are small and learned for one environment. Since 2023, large generative models trained on internet and robot video have been used as interactive simulators of the real world. UniSim [2] learned a simulator from mixed internet, robot and navigation data, and policies trained purely in it transferred to the real world without further training. Navigation World Models (NWM) [3] predict the first-person video of a moving robot and plan by simulating candidate trajectories. NVIDIA's Cosmos platform [4] provides open world foundation models that are meant to be post-trained into world models for particular robots and vehicles.

NVIDIA also builds physics simulators, Isaac Sim and Isaac Lab [12], and uses Cosmos Transfer [13] to make their renderings look real. Cosmos Transfer generates video from spatial control inputs such as depth or segmentation maps, so a simulator rendering can be turned into a real-looking video that keeps its layout and motion, and the simulated actions remain valid. It is the most widely used bridge I have found from simulation data to real-looking data. DreamGen [14] fine-tunes a video world model on the target robot, generates videos of new behaviours, labels them with actions inferred from the video and trains a policy on them. With these data, a humanoid robot learned 22 new behaviours in seen and unseen environments from teleoperation data of a single pick-and-place task, recorded while a human steered the robot.

A newer line merges the world model and the policy. A World Action Model (WAM) is a single network that predicts both future states of the world and the robot's actions. DreamZero [15], a well known model of this kind, reported that it generalized to new tasks and environments more than twice as well as vision-language-action models (VLAs). VLAs map images and instructions directly to actions. The simulation-to-reality transfer of WAMs has barely been studied. The first such work I



have found [7] fine-tuned a Cosmos video model into such a policy, trained it only on synthetic demonstrations and reached 35% average success on a real Franka arm without any real robot demonstrations.

LeCun's position paper [8] argues that a world model should not predict pixels. In his Joint Embedding Predictive Architecture (JEPA), the model encodes the current and the future observation and predicts the future representation, so it can drop unpredictable detail such as texture or sensor noise. DINO-WM [16] predicts the frozen features of DINOv2, a vision network pretrained without labels, and plans without reconstructing pixels. V-JEPA 2 [9] was pretrained on more than a million hours of video. Its action-conditioned version, V-JEPA 2-AC, was further trained on less than 62 hours of robot video from the DROID dataset and planned pick-and-place on real Franka arms in labs it had not seen, without rewards. SkyJEPA [10] trained a JEPA world model on automatically generated data and reported zero-shot simulation-to-reality control of a quadrotor in outdoor flights. In a study of DreamerV3 world models for quadrotor navigation, the model that scored best in simulation failed on the real platform, while the robustness of the learned representation across environments predicted real transfer, and the size of the latent space and the length of the training sequences mattered most for the quality of the world model [17]. A comparison of six encoders as the base of a world model trained on BridgeData V2, a real manipulation dataset, found that encoders trained to reconstruct pixels scored well on pixel metrics, while encoders pretrained on semantic content were better for planning and for evaluating a fixed policy inside the model [18]. In navigation, RAE-NWM [19] predicted DINOv2 features instead of the autoencoder codes of NWM and planned better with them on SCAND but not on RECON. Neither work trained policies inside the models or tested them on real data, and these two steps are what RQ1 adds. I have not found in my search a controlled comparison that measures how the prediction target and the pretrained encoder affect the performance on real data of policies trained inside a world model.

World models are also used to evaluate policies instead of running them on a robot. Ctrl-World [20], trained on DROID, ranked policies without real rollouts and improved a policy by fine-tuning it on imagined successful trajectories. World-in-World [6] tested world models in closed loop on navigation and manipulation and found that visual quality does not guarantee task success, while how well a model follows actions matters more. VLAW [5] found that current world models lack the physical accuracy needed to improve policies, because they are trained on demonstrations without failures and miss small contact details. For physics simulators, the SIMPLER study [21] showed that scenes visually matched to real images can rank policies in nearly the same order as real runs.

Almost all large world models are trained on real video. Training them on physics simulation data is attractive, because such data are cheap, come with exact actions and states and can be generated without limit, but it moves the simulation-to-reality gap from the policy into the world model. The classic tools for this gap were built for policies. Domain randomization [22] varies simulator parameters such as textures so that reality looks like one more variation. Co-training on simulated data together with a small real set improved real-world manipulation [23], and a mechanistic analysis of such co-training showed that it aligns the two domains but keeps them distinguishable [24], so making the two domains identical is not required. For world models, SimDist [25] pretrained a world model in simulation and adapted only the part that predicts the next state with real data, on contact-rich manipulation and quadruped locomotion. Wang et al. [26] pretrained a world model in simulation by imitation and fine-tuned its encoder and policy on a small set of real demonstrations for manipulation, with object poses rather than images as input. SkyJEPA [10] is the clearest case I have found of a model trained without real flight data that worked in reality, on a flight task without contact with objects and with state inputs rather than images. I have not found in my search a controlled study of which representation learning choices, such as the prediction target, appearance randomization or transfer, an objective that pairs simulated and photorealistic views, or co-training, make a world model trained on simulation data predict real recordings well.

When a small amount of real data is available, it is used in different ways. VLAW [5] fine-tunes the world model on real rollouts and then improves the policy with synthetic data from it. DreamGen [14] fine-tunes the video world model on the target robot before generating new training videos. The



Interactive World Simulator [27] reported that policies trained on data from its world model performed comparably to policies trained on the same amount of real data. Parameter-efficient fine-tuning, such as LoRA [28], updates only a small set of added weights and is a common way to adapt large pretrained models cheaply. On the policy side, keeping the visual representation of a VLA close to its pretrained state during fine-tuning improved generalization to new conditions [29], and the best layers to fine-tune were found to depend on the type of distribution shift [30]. These works fix the pretrained representation and vary how it is adapted, and SimDist [25] trained its encoder on simulation data and kept it frozen during real adaptation. I have not found in my search a comparison of how the representation of a world model, and of the policy trained in it, should be pretrained so that the fewest real samples are needed later.

My research questions in §7 address these open points. RQ1 asks which representation learning choices affect how well a policy trained in a world model built from real data performs on real data, RQ2 which choices make a world model trained on simulation data predict the real world, RQ3 how to pretrain the world model and the policy so that a new real setting needs less real data, and RQ4 whether the answers hold in unseen scenes and in navigation.

The works cited in this document are listed in section 12.

7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

The aim of my dissertation is to develop representation learning methods that let learned world models serve as simulators for robot learning that generalizes to the real world. By these I mean the choices of how the world model and the policy learn their internal features: what the world model predicts, which data and encoder they start from, and which objectives make the features ignore what differs between simulated and real images, judged by observable results only.

I will fine-tune existing pretrained models on academic GPU clusters. My main setting is manipulation on BridgeData V2. In navigation I run RQ1 only with the settings chosen on manipulation, without new tuning, and these runs are the test of H4. By performance on real data I mean how closely a policy matches the actions of held-out real recordings and how often it succeeds in SIMPLER, a simulator of scenes rebuilt from BridgeData V2 (section 9). This is a proxy for success on a real robot, which I will test only if access allows.

My research questions (RQ) and hypotheses (H) are as follows.

1. RQ1. When a policy is trained inside a world model built from real data, which representation learning choices affect its performance on real data? I will compare the prediction target of the world model, the encoder the policy is built on and the amount and diversity of the imagined data, and a policy trained directly on the real data that built the world model.
2. H1. With the same predictor and data, policies trained in a world model that predicts the compressed features of an encoder pretrained on real video perform better on real data than those trained in one that predicts the codes of an autoencoder trained to reconstruct pixels. I expect this because such features leave out detail that cannot be predicted, but I test only the observable result. The rival explanation is that performance depends mainly on how faithfully the model follows actions, not on the prediction target. If the gain disappears between models that follow actions equally well, this supports the rival.
3. RQ2. Which representation learning choices make a world model trained on physics simulation data predict real recordings well and give policies whose actions closely match recorded real actions, measured in open loop? In the BridgeData V2 scenes rebuilt in SIMPLER I will compare appearance randomization and photorealistic transfer of simulated frames, randomization of the



simulated physics, an objective that pairs each simulated frame with its photorealistic version, and co-training with a small real set.

4. H2. A world model trained on simulation data transfers to real data better when its encoder is trained to give the same features for a simulated frame and its photorealistic version than when the photorealistic frames are only added to the training set. One rival explanation is that the gap comes mainly from the simulated physics, so that only physics randomization or co-training with real data helps. A second is that photorealistic frames still differ from real ones, so neither use helps.
5. RQ3. How should the encoder of a world model and a policy be pretrained so that adapting them to new real scenes needs less real data? Starting from an encoder pretrained on real video, I will continue its pretraining on real robot data, or on real and simulation data with and without the objective from RQ2, and compare these variants at a few small amounts of real data and several seeds.
6. H3. The encoder further pretrained on real and simulation data with the objective from RQ2 reaches the performance of fine-tuning on all available real data with less real data than the other variants. The rival explanation is that the amount of real data is set by the dynamics of the new scenes rather than by the encoder, so that all variants need the same amount.
7. RQ4. Do the answers hold in unseen scenes, and does the answer to RQ1 hold in navigation with the settings chosen on manipulation and no new tuning? H4. The best choices in manipulation also improve performance on real data over the same baseline in unseen scenes and in navigation, each measured in its own way. The rival explanation is that a gain comes from the data of one task, such as its scene diversity, rather than from the representation choice.
8. Fallback. If these choices make no difference at similar action following, I will instead study the mixing ratios, amount and diversity of generated, simulated and real data.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The main expected contribution of my dissertation is a set of representation learning methods that let learned world models serve as simulators for robot learning that generalizes to the real world. I expect three results. First, evidence on which choices in how a world model and a policy learn their features, such as the prediction target and the pretrained encoder, affect how well a policy trained inside a world model built from real data performs on recorded real data and in SIMPLER, mainly in manipulation and with a check in navigation. Second, training methods for world models built from physics simulation data, in particular objectives that make their features insensitive to the difference between simulated and real appearance, together with evidence on whether randomizing the simulated physics or adding a small real set helps more than changing appearance. Third, pretraining methods that let a world model and a policy be adapted to a new real setting with less real data than full fine-tuning.

World models are becoming a common way to create training data and to evaluate robot policies, NVIDIA and others are combining them with physics simulators, and World Action Models now put the world model and the policy into one network. My results should show which properties of a world model must be right for a given task, which is useful both to those who build world models and to those who train policies in them. Because the questions concern learning under distribution shift, the results on pretraining and invariance objectives may also be useful in domain adaptation more



broadly. I plan to release the code, the evaluation protocol and, where licences allow, the trained models, so that others can compare methods in the same way.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Most of my work will consist of training and fine-tuning world models and policies, measuring how well they transfer to real data and comparing ways of learning their representations. I have not found an established theory that covers the generalization of world models or of policies trained in them, so the work will be mostly empirical, with controlled comparisons in which one choice changes at a time.

I will use PyTorch for training, Git with GitHub repositories for version control and Hugging Face for downloading and publishing models and data. My main setting is manipulation on BridgeData V2, a public dataset recorded with a WidowX robot arm, and my second task is navigation on RECON and SCAND, two real robot datasets on which NWM [3] was trained. For each task I will use one world model that predicts video and one that predicts features, built from the same predictor. For manipulation I will follow the setup of [18], whose code is public, and train the same predictor on the same data to predict either the codes of an image autoencoder trained to reconstruct pixels or the features of an encoder pretrained on real video, such as V-JEPA 2 [9]. For navigation I will do the same with the predictor of RAE-NWM [19], whose code is public, and train a goal-conditioned policy in both models, with only the settings chosen on manipulation and no new tuning. Policies will be trained in a world model by generating imagined trajectories once and fine-tuning the policy on them, rather than by reinforcement learning inside the model, which would be too slow. Ctrl-World [20] kept imagined rollouts that people judged successful and fine-tuned the policy on them. I will instead run a policy trained on real data with added action noise in each world model and keep the rollouts that the same success classifier judges successful. I will respect the licences of the open weights, some of which allow only non-commercial use.

For RQ2 I will generate simulation data with scripted solutions of the tasks in the BridgeData V2 scenes rebuilt in SIMPLER [21], so that the simulated and real scenes match, and add randomized variations of their appearance and physics in ManiSkill, on which SIMPLER is built. If ManiSkill limits the variations I need, I will use version 3.0 of NVIDIA Isaac Lab [12], an early access release as of September 2026. SIMPLER needs graphics drivers that some clusters lack, so in the first weeks I will check that it renders on the cluster, and otherwise I will run it on a workstation GPU. I will drive Cosmos Transfer [13] with depth images to add photorealistic appearance to the renderings. I will transfer a fixed subset of the simulated data once, reuse it for all variants and state its size.

Since I will not have my own robot at the start, I will evaluate on public data. Held-out trajectories of the datasets give real observations, actions and next observations. A world model is tested on them by how well it predicts the recorded next frames. In the first weeks I will check how many BridgeData V2 recordings come from the scenes rebuilt in SIMPLER. In every RQ I will leave them out of world model training and use them as the unseen scenes, and if they are too few, I will use the most similar BridgeData V2 scenes and say so. A policy cannot be run on a recording, so I will measure its performance on real data in two ways. On held-out real trajectories I will measure how closely its actions match the recorded actions. This is open loop and can disagree with success when the policy acts on its own. In SIMPLER the policy acts in closed loop in the rebuilt BridgeData V2 scenes. SIMPLER ranked the policies tested in its study in nearly the same order as real runs, although it was not tested on policies trained in world models. For navigation, where no such simulator exists, I will report how close the path of the policy is to the recorded one and whether its end point reaches the goal, and I will state that this measure rewards imitation of the recorded path. I will call these measures performance on real data. They are not success on a real robot and leave out parts of the



gap, such as the robot's own actuation and camera, and I will state this with every result. If access allows, I will check selected results on a real robot in a university laboratory.

For RQ1 I will train policies inside world models that differ in one choice at a time. The choices are the prediction target, the encoder of the policy and the amount and diversity of the imagined data. In the main comparison the policy reads the representation of the world model it was trained in, and I compare encoders of the policy separately. As a baseline I will train the same policy directly on the real data used to build the world model, and I will report whether imagined data helps beyond it, mainly in SIMPLER, where the policy acts in closed loop. To tell the effect of the prediction target apart from action following, I will check on held-out real transitions whether the prediction made with the recorded action is closer to the recorded next frame than the prediction made with a different action, and report how often the recorded action wins. For feature models I will compare frames from a decoder trained separately from the world model, as in DINO-WM [16], so that the same test applies to every model, and I will report the image quality of every decoder (H1).

For RQ2 I will train world models on the simulation data so that they differ in one choice at a time. The choices are appearance randomization of the simulated frames, photorealistic transfer with Cosmos Transfer, randomization of the simulated physics, a self-supervised objective that gives each simulated frame and its photorealistic version the same features, and co-training with a small set of real trajectories. I will judge each world model by how well it predicts real BridgeData V2 recordings of the matching scenes, with PSNR, LPIPS and FVD, three standard scores of image and video similarity averaged over several samples, and by how closely the policies trained inside it match the recorded actions. I will not use success in SIMPLER as evidence here, because the training data come from the same simulator. For the pairing objective the encoder is fine-tuned, with the frozen encoder as the control. To check the second rival explanation of H2, I will compare the transferred frames with real frames of the matching scenes by the same scores.

For RQ3 I will adapt a world model and a policy to held-out BridgeData V2 scenes with a small set of real trajectories and compare methods at equal amounts of real data. I will continue pretraining only smaller released encoders. The adaptation methods are full fine-tuning, parameter-efficient fine-tuning with LoRA [28] and co-training with generated data. I will first choose the adaptation method with the original encoder and then compare the pretraining variants with that method only. I will repeat this for a few amounts of real data and several seeds and report performance on real data as a function of the amount, which shows how much real data each variant needs to reach the performance of fine-tuning on all available real data (H3). If compute is short, I will drop the variant pretrained on real robot data alone. For RQ4 I will apply the best choices with unchanged settings to held-out scenes and, for RQ1, to navigation. I will call a choice general if it improves performance on real data over the same baseline in every setting, each measured in its own way, and I will not compare the size of the gains across tasks (H4).

For every world model I will report its prediction quality on real recordings. For every policy I will report its performance on real data. I will compare my methods with domain randomization [22], co-training on simulated and real data [23] and the baselines named above, with the same number of training steps and the same amount of real data for every method. To get reliable results I will repeat runs over random seeds and scenes, use statistical tests for comparisons and run ablation studies. Data for training, model selection and final evaluation will be kept separate.

I will run experiments on the H100 nodes of the Wrocław Centre for Networking and Supercomputing (WCSS), with jobs managed by SLURM, and on PLGrid resources. I intend to use AI coding assistants like Claude Code to speed up writing and testing code. They will also help with routine work such as monitoring long experiments and proofreading.

I plan to publish my results at conferences and in journals worth 200 points on the ministerial list. In machine learning these are NeurIPS, ICML and ICLR, in computer vision CVPR, ICCV and ECCV, and in robot learning RSS and the journal IEEE RA-L. Where possible, I will also post the papers as arXiv preprints.



10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Roboty coraz częściej uczą się we „śnie”, czyli w kopii świata przewidywanej przez sieć neuronową zwaną modelem świata, ponieważ nauka w prawdziwym świecie jest powolna, kosztowna i czasem niebezpieczna. Sieć uczy się z nagrań wideo, co zwykle dzieje się po ruchu robota, ale jej sen nigdy nie jest idealny. Robot, który dobrze radzi sobie we śnie, w prawdziwym świecie często zawodzi, zwłaszcza w nowych miejscach. Chcę sprawdzić, jak takie sieci powinny uczyć się swojego wewnętrznego obrazu świata, aby umiejętności ze snu przenosiły się do rzeczywistości. Chcę też zbadać, jak budować takie sny z tanich symulacji fizyki, aby dobrze przewidywały prawdziwy świat, oraz jak przygotować sen i robota, aby do nowego miejsca wystarczyło niewiele prawdziwego doświadczenia. Zbadam głównie chwytanie i przedstawianie przedmiotów, a wyniki sprawdzę także w nawigacji robotów. Efektem mają być metody, dzięki którym roboty będą szybciej, taniej i bezpieczniej uczyć się nowej pracy.

Abstract for general public

Robots are increasingly trained in a "dream", a copy of the world predicted by a neural network called a world model, because real-world learning is slow, costly and sometimes unsafe. The network learns from videos what usually happens after a robot moves, but its dream is never perfect. A robot that does well in the dream often fails in reality, especially in new places. I want to find out how such networks should learn their internal picture of the world so that skills from the dream carry over to reality. I also want to study how to build such dreams from cheap physics simulations so that they predict the real world well, and how to prepare the dream and the robot so that a new place takes only a little real experience. I will study mainly the grasping and moving of objects and check the results in robot navigation. The result should be methods that let robots learn new work faster, more cheaply and more safely.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in



the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

September 2027 (planned submission to a venue worth 200 points on the ministerial list, such as the ICLR 2028 conference or the journal IEEE Robotics and Automation Letters).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

I used Claude (via Claude Code) for literature search, drafting and editing, and I reviewed all content and references myself.

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13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

My supervisor and I plan to meet weekly. At these meetings I will present my progress and recent results, and we will agree on the next steps and discuss our joint papers. I will also attend the weekly meetings of my supervisor's group of PhD students, where we discuss current results, problems and new work in the field. For daily contact we will use email and instant messaging. Code, trained models and the LaTeX sources of papers will be kept in shared Git repositories.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor